The Typing Speed Test Accuracy Checker

Week 1 Assignment - Problem 2 | Java

Problem Summary

Compare equal-length strings position by position, count matching characters, calculate accuracy, and report the first mismatch using a 1-based position.

Java Solution

```
public class Main {
    static void checkTypingAccuracy(String original, String typed) {
        if (original.length() != typed.length()) {
            System.out.println("Inputs must have equal length");
            return;
        }
        int matched = 0, firstMismatch = -1;
        for (int i = 0; i < original.length(); i++) {
            if (original.charAt(i) == typed.charAt(i)) {
                matched++;
            } else if (firstMismatch == -1) {
                firstMismatch = i;
            }
        }
        double accuracy = original.length() == 0 ? 100.0
            : matched * 100.0 / original.length();
        System.out.printf("Matched: %d/%d | Accuracy: %.2f%% | ",
            matched, original.length(), accuracy);
        if (firstMismatch == -1) {
            System.out.println("No Mismatches");
        } else {
            System.out.printf("First Mismatch at position %d ('%c' vs '%c')%n",
                firstMismatch + 1, original.charAt(firstMismatch),
                typed.charAt(firstMismatch));
        }
    }
    public static void main(String[] args) {
        checkTypingAccuracy("hello world", "hello worlt");
        checkTypingAccuracy("coding", "coding");
    }
}
```

Sample Output

```
Matched: 10/11 | Accuracy: 90.91% | First Mismatch at position 11 ('d' vs 't')
Matched: 6/6 | Accuracy: 100.00% | No Mismatches
```

Complexity / Key Notes

Time: O(n). Extra space: O(1). The source specifies equal-length strings; the solution also validates this condition.